

When Additional Demonstrations Hurt: Bayes-Gap Accounting for Shot-Budget Reliability

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Abstract

Language-model systems increasingly make a deployment-time choice about how much context to present before prediction. Extra demonstrations, retrieved evidence, or memory can add task-relevant information, but they also change the prompt distribution through length, order, formatting, separators, and query position. We study this shot-budget reliability problem under log-loss, using nested-shot in-context learning as a controlled case study. The main result is an exact adjacent-step accounting law: the deployed risk change equals negative Bayes gain plus Bayes-gap growth. A harmful log-loss step therefore occurs precisely when excess-risk growth induced by the new prompt exceeds the Bayes information gain supplied by the added context. A second decomposition separates representation deficiency from readout mismatch. The empirical sections turn this accounting law into a practical audit design for black-box LLMs. On ARC-Challenge candidate scoring, the audit exposes a late-shot deterioration case with an observable Bayes-gap floor; order ensembling reduces order-induced mismatch; and held-out readouts recover substantial log-loss from frozen representations, indicating that part of the harm lies in the final prediction rule rather than in unavailable task information. A held-out HellaSwag audit further shows that log-loss-aware shot policies can improve over a fixed high-shot default in model-specific regimes.

Keywords: in-context learning, large language models, log-loss, Bayes gap, prompt sensitivity, shot-budget reliability

1 Introduction

Long-context large language models (LLMs) have turned context into a deployment-time resource. Retrieval systems, decision-support pipelines, and agents decide how many demonstrations, passages, memories, or intermediate traces to place before the query. Many-shot in-context learning (ICL) is the cleanest version of this design problem (Brown et al. 2020; Agarwal et al. 2024; Li et al. 2025; Zou et al. 2025; Chung et al. 2026; Bertsch et al. 2025): the target stays fixed while the prompt grows along a nested shot path. The usual statistical intuition says that relevant additional evidence should help. That intuition is correct for the Bayes predictor, but a deployed LLM receives a serialized prompt. Adding a demonstration also changes length, order, separators, formatting, and query position. When a later shot increases negative log-likelihood (NLL) after earlier gains (Tang et al. 2025; Joo and Klabjan 2025; Rafiei Oskooei et al. 2025; Chen et al. 2023), the useful question is an attribution question: how much of the step is remaining information gain, and how much is new mismatch created by the changed prompt?

We make that attribution exact for log-loss on nested prompt paths. If shot $k+1$ extends shot k by one labeled item, the Bayes predictor cannot be harmed by the extra information. Its entropy term is non-increasing. Any harmful step in the deployed predictor’s log-risk must therefore come from growth in the Bayes gap, the excess log-risk relative to the Bayes predictor. The adjacent-step identity is

$$R_{k+1}(q_\theta) - R_k(q_\theta) = -G_k + \Delta BG_k,$$

where G_k is the one-step Bayes gain and ΔBG_k is one-step Bayes-gap growth. Adding one more shot is harmful exactly when $\Delta BG_k > G_k$ (Fig. 1). Harmful nested-shot prompting is therefore not a mystery about information becoming bad evidence; it is a competition between diminishing information gain and prompt-induced mismatch.

The entropy-divergence decomposition behind the identity is classical (Savage 1971; Gneiting and Raftery 2007; Dawid 2007), and recent statistical theories of ICL also use Bayes-term/Bayes-gap decompositions (Wakayama and Suzuki 2025; Zhang et al. 2025). The contribution here is to difference that decomposition along a fixed test-time path and use the result as a reliability audit. A representation-split law then localizes the mismatch after the prompt is mapped to a representation $Z_k = \phi(\Pi_k)$: a harmful step can reflect Bayes-relevant information missing from the representation, or a readout that is mismatched even though the representation still contains the signal. This links prompt reliability to classical sufficiency and predictive compression (Blackwell 1953; Tishby et al. 2000).

The main claim is deliberately local rather than universal: under log-loss and a nested context path, reliability at the next step is governed by a gain-gap balance. A deployed predictor is helped by the next item only when the remaining Bayes information gain dominates the Bayes-gap growth induced by the new prompt. This matters to machine learning because context length, retrieval depth, and memory exposure are now deployment choices rather than passive inputs; the paper gives a proper-score accounting rule for one such choice.

The paper develops the claim through four contributions. First, Theorem 2 gives the exact one-step gain-gap law, and Corollary 3 extends it to nested context-budget expansion with a fixed prediction target. Second, Theorem 5 separates representation deficiency from readout mismatch, which lets the audit ask whether late-shot harm reflects missing task information or a worse prediction rule on a usable representation. Third, Bayes-invariant prompt perturbations, prompt ensembling, entropy traces, and held-out readouts are connected to conservative observable diagnostics. Fourth, a finite-policy held-out audit turns logged curves into a test of model-specific shot-budget selection under the deployment loss.

The evidence is layered to match those claims. Bayesian reference models make the latent terms computable in solvable families (Fig. 2). ARC-Challenge candidate scoring supplies the main open-model diagnostic setting, because NLL, entropy, and perturbation disagreement are measured on the same cached predictive distributions (Fig. 3). Cross-task rows check breadth (Table 1), mechanism summaries localize where the mismatch sits (Table 2), order ensembling and shift interventions test prompt-form mechanisms (Table 3; Figs. 4 and 5), held-out readouts test readout repairability, and a held-out HellaSwag audit tests whether logged curves can select model-specific shot budgets under log-loss (Table 4); additional ARC splits determine which effects survive protocol changes.

For black-box LLMs, the exact Bayes terms are latent, so the empirical target is the deployed predictor’s observable audit trail. Candidate scoring gives ΔNLL directly. Predictive entropy and entropy drops summarize the gain side of the path, and Bayes-invariant disagreement summarizes sensitivity to prompt form. These observables are diagnostics, not estimates of the hidden terms. Supplementary Appendix H gives the full theory-to-practice dictionary.

The empirical design follows that separation. ARC-Challenge under candidate scoring is the main setting because it produces predictive distributions over a fixed answer set, so NLL, entropy, and perturbation disagreement are measured from the same cached runs. Exact-Bayes reference models make the latent terms computable in solvable families. Open-model experiments then ask three audit questions: whether prompt perturbations move risk in the directions implied by the decomposition, whether a frozen late- k representation still contains recoverable task signal, and whether held-out logged curves can select shot budgets that improve log-loss on disjoint examples.

The object studied here is local and structural: log-loss on nested shot or context-budget paths with a fixed prediction target. The main open-model claims concern the onset grid $k \leq 16$ on fixed evaluation subsets, where candidate scoring gives a clean predictive distribution and controlled prompt perturbations are available. Archived $k=32$ endpoints and broader task checks are reported as stress tests. Non-nested agent trajectories, environment-changing tool actions, best-of- N generation, and learned heuristic controllers require objectives beyond the predictive-risk accounting used here. Within this object, nested-shot ICL is a controlled case study of a general reliability problem: added context can increase information while the induced prompt shift increases conditional mismatch even faster.

Roadmap. Section 2 reviews related work. Section 3 introduces the task, prompt, and context-budget formalization. Section 4 gives the Bayes-gap identities and the

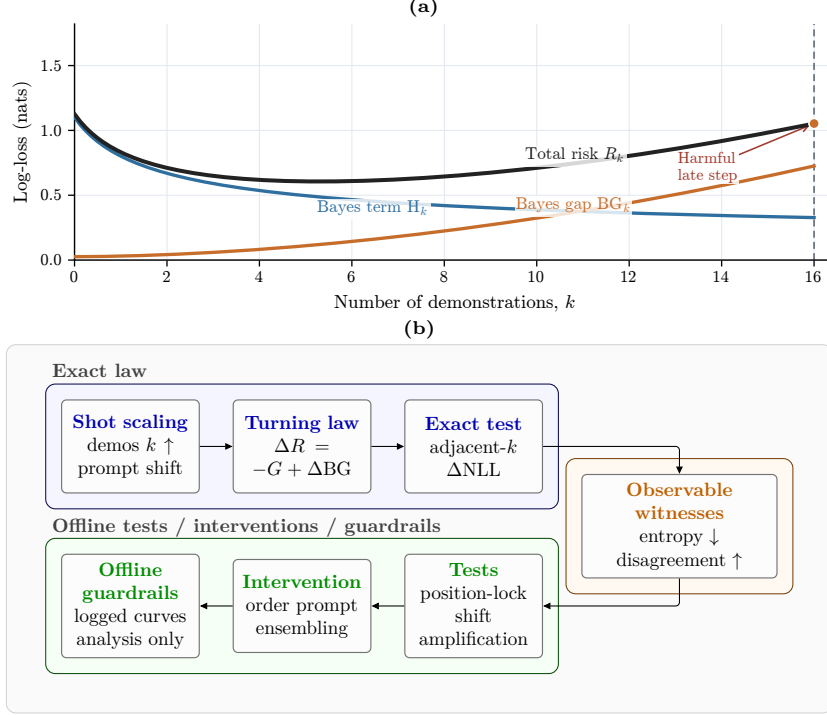


Fig. 1 Harmful shot-scaling steps as Bayes-gap turning points under log-loss. (a) The Bayes term H_k is monotonically non-increasing in k , but the Bayes gap BG_k may grow as the prompt distribution shifts. Their sum can have a harmful adjacent step or, after earlier improvements, a non-monotone turn. (b) The experiments are organized by the decomposition: adjacent- k ΔNLL measures the exact risk change, entropy and disagreement traces are observable proxies whose warning value is quantified separately (Supplementary Appendix I), and Bayes-invariant perturbations together with offline shot-budget analyses audit the prompt-shift mechanism in deployed models (Section 8).

nested context-budget corollary. Sections 5 and 6 add conservative sufficient conditions and observable perturbation diagnostics. Section 7 gives Bayesian reference models, Section 8 reports the open-model diagnostics, and Section 9 states the practical scope. The supplementary material gives limitations, additional experiments, and reproducibility details.

2 Related work

Four lines of work are closest. We organize them around the same attribution question: what extra context contributes, how prompt form can change the deployed predictor, and why deployed systems need a budget-aware reliability view. Many-shot ICL studies motivate the problem: additional demonstrations can become redundant, inefficient, or harmful in particular model-task regimes (Agarwal et al. 2024; Tang et al. 2025; Joo and Klabjan 2025; Rafiei Oskooei et al. 2025; Chen et al. 2023; Zou et al. 2025; Chung et al. 2026; Upasani et al. 2026). That literature shows that shot scaling is

not uniformly beneficial. We study a narrower object inside that behavior: the first harmful log-loss step along a nested prompt path.

Prompt-construction studies explain the mismatch side of the attribution problem. Order, formatting, query placement, and prompt wording can change ICL predictions (Guan et al. 2025; Li et al. 2025; Xiang et al. 2024; Cobbina and Zhou 2025; He et al. 2025; Chatterjee et al. 2024; Razavi et al. 2025). Some of these papers search for better prompt variants. Others treat prompt sensitivity as an evaluation concern. Here those perturbations are interpreted through a loss decomposition: Bayes-invariant changes probe model-side mismatch under prompt shift, and order averaging is used as an intervention motivated by the decomposition rather than as a general prompt-search method.

Bayesian and statistical theories of ICL provide the statistical benchmark. Prior work analyzes implicit Bayesian inference, task inference, Bayes gaps, minimax behavior, and robustness under task shift (Xie et al. 2021; Chan et al. 2022; Wakayama and Suzuki 2025; Zhang et al. 2025; Kim et al. 2024; Ching et al. 2026; Ma et al. 2025; Goddard et al. 2025). Recent empirical work also treats ICL as learning with limited cross-task generality and prompt-dependent robustness (de Wynter 2026). The analysis here uses that viewpoint for harmful shot-scaling onset. It turns the classical log-loss decomposition into a one-step turning law and then separates representation deficiency from readout mismatch.

Long-context reliability work studies related failures such as lost-in-the-middle behavior, length degradation, attention sinks, positional effects, and retrieval-conditioned failures (Liu et al. 2024; Du et al. 2025; Huang et al. 2024; Xiao et al. 2023; Gu et al. 2024; Kazemnejad et al. 2023). These papers support treating shot scaling as a prompt-space distribution shift rather than as a pure increase in sample size. The diagnostic analysis below targets that setting: it asks when additional context remains useful and when the induced shift adds more risk than the added information removes.

Recent LLM systems also treat context and compute as deployment resources. Agentic retrieval, budget-aware context management, and graph-routed advising systems decide which evidence reaches the model (Wu et al. 2026; Singh et al. 2025; Nguyen et al. 2026). Many-shot efficiency and adaptive-shot methods choose or cache demonstrations under per-query cost constraints (Golchin et al. 2025; Xiao et al. 2025; Ou et al. 2026). Semantic-prompt reinforcement learning and test-time scaling methods allocate language guidance or inference computation under stability and latency constraints (Hao et al. 2026; Zhang et al. 2025; Upasani et al. 2026). These papers motivate the context-budget reading of the theory. This paper isolates a local reliability component that such systems inherit whenever added test-time context changes the predictor’s input distribution; controller design, retrieval-policy learning, and heuristic generation are separate system-level problems.

The log-loss focus also places the paper in the literature on proper scoring rules and calibration. Log-loss is a strictly proper score, so the Bayes predictor is the population optimum and the excess risk has a KL form (Savage 1971; Gneiting and Raftery 2007; Guo et al. 2017). This is why the one-step law is an exact accounting identity rather than a fitted diagnostic. The empirical diagnostics are one level weaker: they

ask whether prompt perturbations and calibration splits expose the kind of mismatch that the identity says must explain harmful log-loss steps.

Relative to prior Bayes-gap analyses, the distinction is object and use. The entropy-divergence split is classical, and Bayes-term/Bayes-gap decompositions of ICL risk have been used to quantify how well meta-trained predictors approximate posterior inference across tasks (Wakayama and Suzuki 2025; Zhang et al. 2025). Here the decomposition is differenced along a nested *test-time* path for a fixed deployed predictor. This yields the exact one-step turning criterion and its context-budget extension, an exact Bayes-term cancellation under Bayes-invariant prompt transformations, a representation/readout split for localizing mismatch, and a label-space floor on the Bayes gap under candidate scoring. These are inference-time accounting tools; they are not training-time generalization guarantees.

3 Problem formulation and setup

This section formalizes the attribution problem. ICL is represented as prediction under a hierarchical task distribution, with the abstract evidence (X, D_k) kept distinct from the concrete prompt string Π_k . This separation lets the theory ask whether a harmful step comes from limited added information or from the prompt distribution induced by representing that information. The same notation is used throughout the theory and experiments.

3.1 Hierarchical task model

Let $\mathcal{F}$ be a family of tasks, let $f \sim \rho$ be a random task, and let labeled data follow P_f on $\mathcal{X} \times \mathcal{Y}$ conditioned on f . A k -shot context is $D_k := \{(X_i, Y_i)\}_{i=1}^k$. By convention, $D_0 := \emptyset$, and all quantities such as $R_k(q_\theta)$, H_k , and $\text{BG}_k(q_\theta)$ are defined at $k = 0$ by the same formulas with the empty context. In the canonical hierarchical model used in Bayesian ICL theory, $(X_i, Y_i) \stackrel{i.i.d.}{\sim} P_f$ and the query $(X, Y) \sim P_f$ is drawn independently given f . In practice, however, D_k is often produced by query-dependent retrieval, sorting, or deduplication, so it need not be i.i.d. or independent of X .

Remark 1 (Demonstration selection beyond i.i.d.) The standard i.i.d. within-task model is useful for intuition, but the main identities do not require it. The per- k decomposition identity (1) holds for any joint distribution of (X, D_k, Y) . The Bayes-gain identity $H_k - H_{k+1} = \text{I}(Y; S_{k+1} \mid X, D_k)$ only requires a *nested* shot-scaling path $D_{k+1} = (D_k, S_{k+1})$. Here S_{k+1} denotes the newly added demonstration. It may be selected adaptively and may depend on X (e.g., top- $(k+1)$ retrieval extends top- k by one item). Retrieval-based, sorted, or deduplicated many-shot ICL fits the theory without modification; non-i.i.d. selection only changes the distribution of S_{k+1} . If shot- k and shot- $(k+1)$ prompts use *different* sets of demonstrations, the per- k decomposition still holds, but the one-more-demo interpretation is lost. So is the one-step identity in Theorem 2.

The same nesting condition is useful beyond demonstrations. A test-time system may expose a growing context state Γ_k rather than a set of demonstrations. This state can contain retrieved evidence, memory records, tool traces, or intermediate

summaries, as long as the prediction target and evaluation distribution stay fixed. The main experiments use the many-shot case $\Gamma_k = D_k$, but the accounting below is not tied to demonstrations as objects.

The Bayes reference conditions on the abstract evidence:

$$p^*(\cdot \mid X, D_k) := P(Y \in \cdot \mid X, D_k).$$

A deployed LLM, however, is evaluated through a serialized prompt rather than through direct access to the abstract state.

3.2 Prompt distributions and context budgets

A deployed LLM consumes a *prompt string*, not the abstract pair (X, D_k) . We distinguish the Bayes-relevant variables (X, D_k) from the concrete prompt $\Pi_k = T_k(D_k, X)$ produced by a deterministic template T_k , including ordering conventions. The deployed predictor is therefore written as $q_\theta(\cdot \mid \Pi_k)$. As k changes, the prompt gains labeled information and also changes surface properties such as length, absolute positions, and separators. The random prompt Π_k induces a distribution $\mathcal{P}_k$ on prompt space. Shot scaling therefore changes both the evidence available for prediction and the input distribution seen by the deployed model. The central question is which side dominates at the next step.

4 Bayes-gap analysis for shot scaling

This section gives the formal answer under log-loss. Define the k -shot risk

$$R_k(q_\theta) := \mathbb{E}[-\log q_\theta(Y \mid \Pi_k)].$$

We say that q_θ *overprompts* from k to $k+1$ if $R_{k+1}(q_\theta) > R_k(q_\theta)$.

Definition 1 (Bayes term and Bayes gap) Define the Bayes term (posterior predictive entropy under log-loss)

$$H_k := \mathbb{E}[-\log p^*(Y \mid X, D_k)] = H(Y \mid X, D_k),$$

and the Bayes gap

$$\text{BG}_k(q_\theta) := \mathbb{E}[\text{KL}(p^*(\cdot \mid X, D_k) \parallel q_\theta(\cdot \mid \Pi_k))].$$

When the predictor q_θ is fixed throughout a discussion, we sometimes abbreviate $\text{BG}_k(q_\theta)$ as BG_k .

The quantity $\text{BG}_k(q_\theta)$ compares the abstract-evidence Bayes posterior $p^*(\cdot \mid X, D_k)$ with the deployed predictor on the serialized prompt Π_k , so it already absorbs information lost in serialization as well as readout mismatch. Theorem 5 and Corollary 6 make this split explicit; BG_k is pure readout mismatch only when the serialization is Bayes-lossless.

The entropy-divergence split for the logarithmic score is classical (Cover and Thomas 2006; Savage 1971; Gneiting and Raftery 2007; Dawid 2007). Here it identifies

the source of non-monotonicity under log-loss and yields an exact one-step criterion for the attribution question. Proofs of the formal statements in this and the next two sections are collected in Supplementary Appendix C.

Lemma 1 (Classical decomposition and one-step Bayes gain) *For every k ,*

$$R_k(q_\theta) = H_k + BG_k(q_\theta). \quad (1)$$

If the shot-scaling path is nested, $D_{k+1} = (D_k, S_{k+1})$, then

$$H_k - H_{k+1} = I(Y; S_{k+1} | X, D_k) \geq 0. \quad (2)$$

Under the stated nesting condition, both identities hold for any joint distribution of (X, D_k, S_{k+1}, Y) ; no i.i.d. or within-task assumptions are required.

Theorem 2 (One-step gain-gap criterion) *Fix any predictor q_θ and any nested shot-scaling path $D_{k+1} = (D_k, S_{k+1})$. Define the one-step Bayes gain and one-step Bayes-gap growth by*

$$G_k := H_k - H_{k+1}, \quad \Delta BG_k := BG_{k+1}(q_\theta) - BG_k(q_\theta).$$

Then

$$R_{k+1}(q_\theta) - R_k(q_\theta) = -G_k + \Delta BG_k. \quad (3)$$

Equivalently,

$$R_{k+1}(q_\theta) > R_k(q_\theta) \iff \Delta BG_k > G_k.$$

Under log-loss, overprompting can begin only when one-step Bayes-gap growth overtakes one-step Bayes gain.

Corollary 3 (Nested context-budget paths) *Let $\Gamma_0, \Gamma_1, \dots$ be any nested sequence of test-time context states with $\Gamma_{k+1} = (\Gamma_k, V_{k+1})$, where V_{k+1} is the newly exposed context item. For a fixed prediction target Y and query X , define*

$$R_k(q_\theta) := \mathbb{E}[-\log q_\theta(Y | \Pi_k)], \quad H_k := H(Y | X, \Gamma_k),$$

and define $BG_k(q_\theta)$ by replacing D_k with Γ_k in the Bayes-gap definition. Then

$$R_{k+1}(q_\theta) - R_k(q_\theta) = -I(Y; V_{k+1} | X, \Gamma_k) + BG_{k+1}(q_\theta) - BG_k(q_\theta).$$

Thus the gain-gap law applies to any nested test-time context expansion under log-loss. It covers many-shot prompting as a special case, and it also covers passive expansions such as adding retrieved evidence or stored memory. It does not cover actions that change the environment, non-nested replacement policies, or selection rules that choose among several generated outputs.

Corollary 4 (Coarse-grid implication) *For any $k < k'$, the cumulative change over the interval telescopes as*

$$R_{k'}(q_\theta) - R_k(q_\theta) = - \sum_{j=k}^{k'-1} G_j + \sum_{j=k}^{k'-1} \Delta BG_j. \quad (4)$$

If a coarse grid step is harmful, $R_{k'}(q_\theta) > R_k(q_\theta)$, then there exists at least one intermediate index $j \in \{k, \dots, k' - 1\}$ such that

$$\Delta BG_j > G_j.$$

Equivalently, a harmful adjacent-grid step means that a one-step turning point occurred somewhere inside the interval.

Theorem 5 (Representation/readout split) *Fix k and let $S_k := (X, D_k)$ denote the full task state. Let $Z_k = \phi(\Pi_k)$ be any representation of the prompt. Since Π_k is a deterministic function of S_k , Z_k is a coarsening of S_k . Assume the predictor factors through this representation, so that $q_\theta(\cdot \mid \Pi_k) = q_\theta(\cdot \mid Z_k)$. This factoring holds exactly when Z_k is the final-layer query hidden state used by the held-out readout experiment; the observable metadata map in Section 5 is used only to estimate prompt shift. Write $p^*(\cdot \mid Z_k)$ for the Bayes posterior after observing only the representation Z_k . Define*

$$A_k(\phi) := \mathbb{E}[\text{KL}(p^*(\cdot \mid S_k) \parallel p^*(\cdot \mid Z_k))], \quad C_k(q_\theta; \phi) := \mathbb{E}[\text{KL}(p^*(\cdot \mid Z_k) \parallel q_\theta(\cdot \mid Z_k))].$$

Then the k -shot log-loss risk admits the exact decomposition

$$R_k(q_\theta) = H(Y \mid S_k) + A_k(\phi) + C_k(q_\theta; \phi).$$

Also,

$$A_k(\phi) = I(Y; S_k \mid Z_k),$$

so $A_k(\phi)$ is exactly the Bayes-relevant information discarded by the representation, while $C_k(q_\theta; \phi)$ is the residual fit/readout mismatch after conditioning on Z_k . Along any nested shot-scaling path,

$$R_{k+1}(q_\theta) - R_k(q_\theta) = -G_k + \Delta A_k + \Delta C_k,$$

where $G_k := H(Y \mid S_k) - H(Y \mid S_{k+1})$, $\Delta A_k := A_{k+1}(\phi) - A_k(\phi)$, and $\Delta C_k := C_{k+1}(q_\theta; \phi) - C_k(q_\theta; \phi)$.

Corollary 6 (Lossless full-prompt law) *Take ϕ to be the identity map on the full prompt, so that $Z_k = \Pi_k$. Assume the serialization is Bayes-lossless on the evaluation support:*

$$p^*(\cdot \mid \Pi_k) = p^*(\cdot \mid S_k) \quad \text{almost surely.}$$

Then $A_k(\phi) = 0$ for all k , $C_k(q_\theta; \phi) = \text{BG}_k(q_\theta)$, and Theorem 5 reduces to Theorem 2. Without this losslessness condition, the full prompt may itself be a coarsening of S_k , and the residual representation-deficiency term is $I(Y; S_k \mid \Pi_k)$.

Lemma 7 (Same-representation comparisons isolate readout mismatch) *Fix k and a representation $Z_k = \phi(\Pi_k)$. Let q_1 and q_2 both factor through Z_k , so $q_i(\cdot \mid \Pi_k) = q_i(\cdot \mid Z_k)$ for $i \in \{1, 2\}$. Then*

$$R_k(q_1) - R_k(q_2) = C_k(q_1; \phi) - C_k(q_2; \phi).$$

In particular, when the representation is held fixed, any held-out reduction in log-loss comes entirely from reducing the readout-mismatch term.

The full-prompt law identifies why overprompting can occur under log-loss. The representation split then asks where the failure enters. In the language of Blackwell sufficiency, $A_k(\phi)$ is the Bayes-relevant information discarded by the representation, while $C_k(q_\theta; \phi)$ is the residual prediction error once the representation has been fixed (Blackwell 1953; Tishby et al. 2000). The exact-Bayes reference models vary A_k and C_k in a tractable family. By Lemma 7, improvements from a held-out readout probe reduce only the readout-mismatch term on a fixed hidden-state representation.

Empirical content. Theorem 2 and Corollary 4 locate every harmful step at a crossing $\Delta \text{BG}_k > G_k$ of two unobserved terms. This is exact but, taken alone, unfalsifiable: it holds for any predictor and any data. The identity also does not imply any particular

pattern in observable proxies; whether entropy drops and Bayes-invariant disagreement growth inherit the crossing structure is an empirical question about proxy fidelity, and the answer can differ across model families. The falsifiable content lies one level up, in the prompt-shift reading of ΔBG_k : interventions that amplify prompt shift at fixed evidence should worsen harmful regimes, interventions that average over Bayes-invariant prompt realizations should improve the transformed-family average (Proposition 13), and removing a single geometric factor need not suffice. Section 8.3 tests these interventions; Sections 8.2 and 8.4 then measure, rather than assume, the warning value of the proxies.

Remark 2 (The Bayes predictor cannot overprompt) If the deployed predictor inverts the serialization, i.e. $q_\theta(\cdot \mid \Pi_k) = p^*(\cdot \mid X, D_k)$ almost surely for every realization and every k , then $BG_k(q_\theta) = 0$ for all k , so $R_k(q_\theta) = H_k = H(Y \mid X, D_k)$ is non-increasing in k . Any observed overprompting under log-loss means that the deployed model’s Bayes gap changes with the prompt distribution induced by the template and shot-scaling path.

Remark 3 (An observable Bayes-gap floor) Under candidate scoring over a finite answer set $\mathcal{Y}$, $H_k \leq \log |\mathcal{Y}|$, so the decomposition (1) yields a fully observable, label-based lower bound

$$BG_k(q_\theta) \geq R_k(q_\theta) - \log |\mathcal{Y}|.$$

Whenever measured NLL exceeds $\log |\mathcal{Y}|$ (about 1.386 nats for four candidates), the Bayes gap is certifiably positive, with no proxy, perturbation, or stability assumption involved. The bound discards the unknown slack between H_k and $\log |\mathcal{Y}|$, but it is directly measurable and binds in our data (Section 8.2).

The Bayes term can only improve with additional labeled context, but the marginal gain G_k naturally saturates once the next demonstration is nearly redundant given (X, D_k) . In that regime, even modest prompt-induced Bayes-gap growth can create a turning point. In regular Bayesian models, such marginal gains often decay with k , for example through classical redundancy or mutual-information asymptotics (Clarke and Barron 1990; Barron et al. 1998; Krause et al. 2008).

Remark 4 (Log-loss versus accuracy) Many ICL benchmarks report accuracy or exact match, while our core decomposition is stated under log-loss because it yields an exact Bayes term/Bayes gap split. Log-loss and accuracy can decouple (e.g., calibration can degrade without changing argmax predictions), but Bayes gaps still control worst-case zero-one regret through standard probability-metric inequalities. Supplementary Appendix E collects these links, showing how log-loss and the entropy/sensitivity diagnostics can give early warning before an accuracy drop.

5 Checkable sufficient conditions

The previous section gives exact identities. This section gives conservative certificates that can be checked from prompt representations. Shot scaling is treated as distribution shift in prompt space, and the question is when an explicit stability budget is

small enough for information gain to dominate prompt-induced instability. The certificates are sufficient conditions, not characterizations of every turning point; in the open-model audit, their numerical thresholds are below observed hidden-state drifts (Supplementary Appendix G), so they serve as stability requirements rather than empirical approvals. The stability assumption below is adapted from optimal transport and Wasserstein distributional robustness (Villani 2009; Peyré and Cuturi 2019; Mohajerin Esfahani and Kuhn 2018; Wakayama and Suzuki 2025).

5.1 Representations and Wasserstein distances

Let ϕ map a prompt into a representation space $\mathcal{Z}$. It may be an observable feature map, such as query absolute position, demonstration-block length, and separator or format-token indicators, or an internal representation such as the final-layer query hidden state. We focus on two choices used later. (i) An *observable* metadata map makes the per-shot shift Δ_k directly estimable and clarifies that Δ_k need not vanish as k grows (Remark 5). (ii) An *internal* representation map ties stability to local Jacobian/operator-norm behavior and supports open-model auditing (Supplementary Appendix G). All results hold for any ϕ ; the choice trades interpretability and ease of checking against tightness. Let μ_k be the distribution of $\phi(\Pi_k)$ when (f, D_k, X) is sampled from the hierarchical model. Let $W_1(\mu, \nu)$ denote the 1-Wasserstein distance over $\mathcal{Z}$.

We write $\Delta_k := W_1(\mu_{k+1}, \mu_k)$ for the per-shot shift induced by adding one more demonstration under the fixed template.

In practice, $\Delta_k = W_1(\mu_{k+1}, \mu_k)$ can be estimated from samples of prompt representations $\phi(\Pi_k)$, although Wasserstein estimation becomes harder in high dimensions (Fournier and Guillin 2015). This makes low-dimensional prompt variables, such as query position and prompt length, useful in long-context audits.

Assumption 8 (Wasserstein stability of the Bayes gap) *There exists $L_{\text{BG}} > 0$ such that for all k ,*

$$\text{BG}_{k+1}(q_\theta) - \text{BG}_k(q_\theta) \leq L_{\text{BG}} W_1(\mu_{k+1}, \mu_k).$$

Assumption 8 is one-sided: it upper-bounds only Bayes-gap growth, the direction needed for a safe-improvement certificate. A sufficient route is to write the Bayes gap as the expectation of an L_{BG} -Lipschitz functional of $Z_k = \phi(\Pi_k)$; Kantorovich-Rubinstein duality then gives the bound, as in standard Wasserstein-DRO and domain-shift analyses (Villani 2009; Peyré and Cuturi 2019; Mohajerin Esfahani and Kuhn 2018; Bousquet and Elisseeff 2002; Ben-David et al. 2010). Supplementary Appendix G, Proposition 2, states this route formally. The required functional must be stable across k on the chosen representation. This is natural for Bayes-sufficient representations (Section 7.2); for lossy metadata or hidden-state maps it becomes an additional sequence-level assumption, because the conditional law of the task state given the representation can itself drift with prompt growth.

5.2 Conservative improvement condition

The exact criterion in Theorem 2 is not directly checkable for black-box LLMs because it involves p^* . Assumption 8 replaces the unknown Bayes-gap growth with a stability constant L_{BG} and the observable prompt shift Δ_k , giving a sufficient condition for improvement.

Theorem 9 (Conservative improvement theorem) *Under Assumption 8, if $G_k > L_{\text{BG}} \Delta_k$, then $R_{k+1}(q_\theta) < R_k(q_\theta)$. In particular, the step $k \rightarrow k+1$ is safe under this sufficient condition whenever $G_k > L_{\text{BG}} \Delta_k$.*

Theorem 9 is one-sided. Once $G_k \leq L_{\text{BG}} \Delta_k$, improvement may still occur, but the sufficient condition no longer certifies it; in that case, overprompting becomes possible and instability diagnostics become informative.

Open models. For open models, Supplementary Appendix G develops model-side local audits based on paired prompt representations, output-projection norms, and hidden-state drift measures. These quantities do *not* by themselves identify the total Bayes-gap growth term. They indicate when model-side sensitivity may exceed the level allowed by the sufficient condition.

Remark 5 (Prompt shift need not vanish) Even with i.i.d. demonstrations, increasing k changes prompt geometry: the query moves to larger absolute positions and separator/format tokens shift. If $\phi(\Pi)$ includes the (scaled) absolute query position and the template moves the query by $\ell > 0$ tokens when adding one demonstration, then $\Delta_k \geq \ell$ deterministically, so once Bayes gains decay the safe-improvement condition must eventually fail unless stability improves with length.

Proposition 10 (Exit from the conservative condition under nonvanishing shift) *Suppose that for all $k \geq k_0$ the marginal Bayes gain obeys $G_k \leq a/(k+b)^\alpha$ for some $a, b, \alpha > 0$, and the per-shot prompt shift is bounded below by $\Delta_k \geq \delta > 0$. Then the safe-improvement condition of Theorem 9 must fail for all sufficiently large k , in particular once*

$$k \geq \max \left\{ k_0, \left(\frac{a}{L_{\text{BG}} \delta} \right)^{1/\alpha} - b \right\}.$$

Hence beyond this scale the sufficient condition no longer rules out overprompting.

In many Bayesian tasks the marginal gains G_k do decay with k (often polynomially); for example, in the hierarchical Gaussian mean model of Section 7.1, $G_k = \Theta(k^{-2})$.

5.3 Local Bayes-gap sensitivity audits

Theorem 9 shows that improvement follows when Bayes gain dominates the instability budget $L_{\text{BG}} \Delta_k$. Assumption 8 summarizes sensitivity to shot-induced prompt-space shift through one unobserved constant, L_{BG} . Supplementary Appendix G does not

identify that total constant directly; it develops model-side local sensitivity measures that become sufficient only after extra Bayes-side smoothness assumptions are added. Standard dot-product self-attention is not globally Lipschitz (Kim et al. 2021), so a *global* L_{BG} can be infinite without additional restrictions. Useful *local* bounds can still hold on typical sets and can depend on sequence length and attention-score statistics (Castin et al. 2024; Yudin et al. 2025). For this paper, the implication is limited but useful: stability constants can be bounded or regularized for open models, and effective L_{BG} can worsen with context length, making harmful shot scaling more plausible in long prompts.

For open models, Supplementary Appendix G gives local, data-dependent audits based on hidden-state drift, operator-norm estimates, and paired-prompt shift proxies. These quantities do not identify total Bayes-gap growth by themselves, but they state what a conservative stability certificate would require. In our open-model audit, a necessary condition for the certificate of Theorem 9 is $\hat{\Delta}_k^{\text{cpl}} < \log 4 / \hat{L}_{\text{mdl}}$, where $\hat{\Delta}_k^{\text{cpl}}$ is the empirical mean paired-prompt hidden-state drift and $\hat{L}_{\text{mdl}}$ is the model-side readout Lipschitz surrogate (both defined in Supplementary Appendix G); this corresponds to final-layer drifts below 0.005–0.057 in Euclidean norm depending on the model (Supplementary Appendix G, Table S12). The observed drifts are much larger, so the certificate approves no empirical grid step. The same audit reports an anecdotal three-model ordering between the model-side norm and observed shot-scaling fragility; this observation motivates the logged-curve diagnostics without carrying a main empirical conclusion.

Estimating Jacobian/operator norms can be expensive for large LLMs, so these bounds are most useful as offline or periodic audits for a fixed model/template family. The lightweight stopping heuristic in Eq. (5) instead uses only forward passes through entropy and prompt-disagreement proxies. In the hidden-state construction in Supplementary Appendix G, $\sqrt{2}\|W_{\text{out}}\|_{\text{op}}$ is directly checkable once per model snapshot; converting it into a bound on total Bayes-gap growth still requires Bayes-side smoothness assumptions.

Two simple examples. Supplementary Appendix D gives two short examples showing that overprompting does not require mislabeled or irrelevant demonstrations: a threshold out-of-distribution length construction and a smoother coverage-dilution tradeoff with an explicit one-step condition.

6 Prompt perturbations and interventions

Theorem 2 isolates Bayes-gap growth as the only source of non-monotonicity under log-loss, but BG_k is unobserved for LLMs. This section turns the attribution question into observable diagnostics. Bayes-invariant perturbations keep the abstract evidence fixed while changing the prompt surface seen by the deployed model. The same perturbation families are then used for directly testable interventions. The central object is *prompt sensitivity*, defined as changes in expected log-loss under intent-preserving prompt transformations (Chatterjee et al. 2024; Razavi et al. 2025; Guan et al. 2025; Ma et al. 2025). Under Bayes-invariant transformations, this sensitivity is controlled by Bayes gaps.

6.1 Intent-preserving prompt transformations

Let $\mathcal{G}$ be a set of measurable transformations on prompts (template paraphrases, formatting edits, demonstration re-orderings, etc.). Let $g \in \mathcal{G}$ act on the prompt string: $\Pi_k \mapsto g(\Pi_k)$.

Definition 2 (Bayes-invariant transformations) A prompt transformation g is Bayes-invariant on the evaluation support if it changes only the serialization seen by the deployed model and leaves the abstract evidence for the target unchanged. Formally, if $\Pi_k = T_k(D_k, X)$ and $g(\Pi_k)$ is another serialization of the same abstract state (X, D_k) , then

$$p^*(\cdot \mid X, D_k; g) = p^*(\cdot \mid X, D_k),$$

where the left-hand side denotes the Bayes posterior under the transformed prompt view. We call a family $\mathcal{G}$ Bayes-invariant when this equality holds for every $g \in \mathcal{G}$ and every evaluated realization.

Remark 6 (Strong vs. weak invariances) Bayes invariance is defined relative to the *evaluation loss* and to what is treated as abstract evidence. In the order-invariance experiments, D_k denotes the unordered multiset of demonstrations; re-ordering acts only on the serialization map T_k , not on the evidence itself. Under that convention, demonstration re-ordering, padding-based query-position shifts, and purely syntactic separator edits are strong invariances for label-based candidate scoring. If a data-generating process or evaluation protocol makes demonstration order itself informative, then order permutations are not Bayes-invariant and should be treated only as robustness checks. Likewise, instruction or demonstration paraphrases can change the induced label distribution under rigid string matching, even if they would be equivalent under a semantics-aware judge; they are conditionally invariant and are not used as primary evidence here (Hua et al. 2025).

6.2 Expected log-loss sensitivity

Let ν be a distribution over $\mathcal{G}$ (e.g., uniform over a library of perturbations). Define the transformed risk

$$R_k^g(q_\theta) := \mathbb{E}[-\log q_\theta(Y \mid g(\Pi_k))],$$

and the expected absolute sensitivity

$$\text{Sens}_k^{\text{loss}}(q_\theta; \nu) := \mathbb{E}_{g \sim \nu} |R_k^g(q_\theta) - R_k(q_\theta)|.$$

Theorem 11 (Exact Bayes-term cancellation under Bayes-invariant transforms) *Assume $\mathcal{G}$ is Bayes-invariant. Then for every $g \in \mathcal{G}$,*

$$R_k^g(q_\theta) - R_k(q_\theta) = \text{BG}_k^g(q_\theta) - \text{BG}_k(q_\theta),$$

where $\text{BG}_k^g(q_\theta) := \mathbb{E}[\text{KL}(p^*(\cdot \mid X, D_k) \parallel q_\theta(\cdot \mid g(\Pi_k)))]$ is the Bayes gap under transformed prompts. For any distribution ν over $\mathcal{G}$,

$$\text{Sens}_k^{\text{loss}}(q_\theta; \nu) \leq \text{BG}_k(q_\theta) + \mathbb{E}_{g \sim \nu} \text{BG}_k^g(q_\theta).$$

Since both $\text{BG}_k(q_\theta)$ and $\mathbb{E}_{g \sim \nu} \text{BG}_k^g(q_\theta)$ are nonnegative, Theorem 11 implies

$$\max \left\{ \text{BG}_k(q_\theta), \mathbb{E}_{g \sim \nu} \text{BG}_k^g(q_\theta) \right\} \geq \frac{1}{2} \text{Sens}_k^{\text{loss}}(q_\theta; \nu).$$

Large prompt sensitivity in expected log-loss implies a large Bayes gap under at least one of the original or transformed prompt distributions.

A disagreement check for Bayes-gap smallness. The sensitivity $\text{Sens}_k^{\text{loss}}$ still requires labels. In candidate-scoring settings, however, we can compare $q_\theta(\cdot \mid \Pi_k)$ and $q_\theta(\cdot \mid g(\Pi_k))$ directly, without access to Y . Under Bayes invariance, large disagreement is impossible if *both* prompts are close to the Bayes predictor. The next proposition states this constraint using a metric.

Proposition 12 (Disagreement diagnostic via total variation) *Assume g is Bayes-invariant. Fix a realization (X, D_k) and let $p = p^*(\cdot \mid X, D_k)$, $q = q_\theta(\cdot \mid \Pi_k)$, and $q^g = q_\theta(\cdot \mid g(\Pi_k))$. Let $d_{\text{TV}} := \text{TV}(q, q^g) = \frac{1}{2} \|q - q^g\|_1$. Then*

$$\max\{\text{KL}(p\|q), \text{KL}(p\|q^g)\} \geq \frac{1}{2} d_{\text{TV}}^2.$$

In particular, if both Bayes gaps are $\leq \varepsilon$, then $d_{\text{TV}} \leq \sqrt{2\varepsilon}$.

Necessary-condition checks are clearest for *metrics* on predictive distributions. Proposition 12 is stated in total variation because the proof uses only triangle inequality and Pinsker. KL disagreement, which is reported in the experiments for continuity with prior prompt-sensitivity work, is asymmetric and not itself a metric. When a metric diagnostic is needed, TV, or another true metric such as $\sqrt{\text{JS}}$, should be logged alongside KL.

Pointwise,

$$\text{KL}(p\|q) + \text{KL}(p\|q^g) \geq \max\{\text{KL}(p\|q), \text{KL}(p\|q^g)\} \geq \frac{1}{2} d_{\text{TV}}^2.$$

Taking expectations over (X, D_k) (and optionally over $g \sim \nu$) gives an observable lower bound on the *sum* of Bayes gaps:

$$\text{BG}_k(q_\theta) + \mathbb{E}_{g \sim \nu} \text{BG}_k^g(q_\theta) \geq \frac{1}{2} \mathbb{E}_{g \sim \nu} \mathbb{E}[\text{TV}(q, q^g)^2].$$

In particular, $\max\{\text{BG}_k(q_\theta), \mathbb{E}_{g \sim \nu} \text{BG}_k^g(q_\theta)\} \geq \frac{1}{4} \mathbb{E}_{g \sim \nu} \mathbb{E}[\text{TV}(q, q^g)^2]$.

Prompt ensembling as an intervention. Bayes-invariant prompt transformations also suggest an intervention: average predictive distributions across $g(\Pi_k)$ rather than committing to a single order or template realization. Under log-loss, that intervention admits a standard Jensen bound.

Proposition 13 (Bayes-invariant prompt ensembling reduces average log-loss) *Assume $\mathcal{G}$ is Bayes-invariant and let ν be a distribution on $\mathcal{G}$. Define the ensemble predictor by*

$$\bar{q}_\nu(\cdot \mid \Pi_k) := \mathbb{E}_{g \sim \nu} [q_\theta(\cdot \mid g(\Pi_k))].$$

Then

$$R_k(\bar{q}_\nu) \leq \mathbb{E}_{g \sim \nu} R_k^g(q_\theta), \quad \text{BG}_k(\bar{q}_\nu) \leq \mathbb{E}_{g \sim \nu} \text{BG}_k^g(q_\theta).$$

Bayes-invariant ensembling cannot increase the transformed-family average log-loss, and weakly reduces the corresponding average Bayes gap.

Proposition 13 gives the formal basis for the order-ensemble intervention: averaging across Bayes-invariant orderings weakly reduces the transformed-family Bayes gap under log-loss (Mizrahi et al. 2024; Voronov et al. 2024).

Remark 7 (Scope of Proposition 13) The proposition bounds the ensemble against the transformed-family average $\mathbb{E}_{g \sim \nu} R_k^g(q_\theta)$; it does not bound the ensemble against the base prompt, which is a single member of the family. The ensemble-minus-base improvements reported in Table 3 are therefore empirical findings rather than corollaries: they indicate that, for these models, the base ordering enjoys no systematic advantage over Bayes-invariant re-orderings.

Choosing k from observable proxies. We approximate the gain-gap criterion through label-free proxies: let $\hat{G}_k$ be the entropy drop and let $\widehat{\text{Sens}}_k$ be a Bayes-invariant disagreement proxy, for example TV/JS/KL between $q_\theta(\cdot \mid \Pi_k)$ and $q_\theta(\cdot \mid g(\Pi_k))$. In the main experiments we report KL-based disagreement for continuity with prior prompt-sensitivity work, while Proposition 12 shows that a metric disagreement such as TV yields a clean necessary-condition diagnostic for Bayes-gap smallness.

Remark 8 (The entropy proxy is confounded at large k) The gain-side proxy $\hat{H}_k := \mathbb{E}[\mathcal{H}(q_\theta(\cdot \mid \Pi_k))]$ coincides with the Bayes term only in the calibrated limit $q_\theta = p^*$, in which case it is non-increasing along any nested path. A sustained rise of $\hat{H}_k$ —visible at large k in Fig. 3—is therefore impossible for a calibrated predictor and is itself a symptom of model-posterior decoupling rather than information about G_k . The proxy thus plays two roles along the path: early, a falling $\hat{H}_k$ is read as diminishing-gain evidence; late, a rising $\hat{H}_k$ is mismatch evidence. Rule (5) remains operationally well-defined in both regimes—a non-positive entropy drop triggers stopping either way—but interpretations of the curves must not treat late- k entropy movements as statements about the Bayes term.

A simple sequential rule is to stop at the first k such that

$$\hat{G}_k \leq \lambda \widehat{\text{Sens}}_k, \tag{5}$$

with λ calibrated offline (Supplementary Appendix I). The coefficient λ is a heuristic trade-off constant, not a theorem-derived quantity; Supplementary Appendix I, Table S15, reports sensitivity to this choice. In the experiments, risk or accuracy is plotted against k together with entropy and disagreement proxies. The predicted turning region is where entropy drops flatten while disagreement rises, but the predictive value of those proxies is measured empirically rather than assumed.

Finite policy audits. The same offline view covers finite classes of predeclared shot policies, such as fixed budgets and locked uncertainty selectors. Let $\mathcal{A}$ be such a finite class. Each policy maps a logged example to a candidate predictive distribution. For a numerical floor $\tau > 0$, define the clipped log-loss $\ell_\tau(a; X, Y) := -\log \max\{a_Y(X), \tau\}$, so $0 \leq \ell_\tau \leq L_\tau := \log(1/\tau)$. If $\hat{a}$ minimizes empirical clipped log-loss on an independent calibration set of size n , then, with probability at least $1 - \delta$,

$$R_\tau(\hat{a}) \leq \min_{a \in \mathcal{A}} R_\tau(a) + 2L_\tau \sqrt{\frac{\log(2|\mathcal{A}|/\delta)}{2n}}. \quad (6)$$

This is the standard finite-class uniform-convergence bound from Hoeffding’s inequality and a union bound (Hoeffding 1963). It justifies the offline protocol used here: select from a small, predeclared policy class on a calibration split, then evaluate once on a disjoint test split.

7 Bayesian reference models

The laws above are model-agnostic. We use tractable Bayesian reference models as a calibration point for the central claim: when the Bayes term can be computed exactly, the turning point does not come from information becoming harmful, but from model-side mismatch overtaking diminishing gain. In these reference models, Bayes gains, Bayes-gap growth, and the representation/readout split can be computed rather than inferred from proxies. Conjugate Gaussian models are standard for this purpose (Bishop 2006; Rasmussen and Williams 2006) and connect naturally to recent transformer theories of in-context regression (Kim et al. 2024; Ching et al. 2026). The analysis provides reference cases for interpreting the empirical measurements on open models.

7.1 Scalar Gaussian mean model with length shift

Consider a scalar hierarchical regression task with $\theta \sim \mathcal{N}(0, 1)$ and $Y_i \mid \theta \sim \mathcal{N}(\theta, \sigma^2)$. Given $D_k = \{Y_i\}_{i=1}^k$, the Bayes predictive is Gaussian with mean scaling $a_k^* = \frac{k}{k+\sigma^2}$ applied to the sample mean and predictive variance $\sigma^2 + v_k$ with $v_k = \frac{\sigma^2}{\sigma^2+k}$. Hence $H_k = \frac{1}{2} \log(2\pi e(\sigma^2 + v_k))$ decreases in k .

To model length-generalization failure, let a learned predictor become under-confident beyond a training range $k \leq k_{\text{train}}$ by replacing a_k^* with

$$\hat{a}_k = a_k^* \exp(-(k - k_{\text{train}})/\tau) \quad (k > k_{\text{train}}).$$

Under equal predictive variances, the Bayes gap is an expected Gaussian KL,

$$\text{BG}_k = \mathbb{E}[(\hat{a}_k - a_k^*)^2 \bar{Y}^2] / (2(\sigma^2 + v_k)),$$

so Theorem 2 is exactly testable. This first reference model isolates a pure length-generalization failure: the Bayes term remains non-increasing while the learned predictor becomes increasingly miscalibrated beyond its training range. Fig. 2 reports

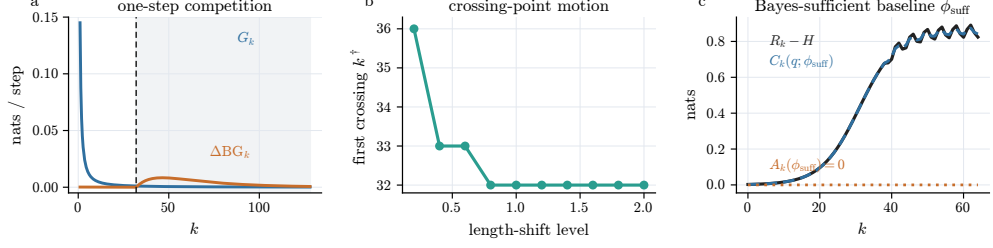


Fig. 2 Bayesian reference models. **(a)** Exact one-step competition in the scalar Gaussian mean model: Bayes gain G_k and Bayes-gap growth ΔBG_k cross at the turning point. **(b)** Crossing-point motion under a controlled level of out-of-distribution length degradation: stronger post-training degradation shifts the first crossing to smaller k . **(c)** Bayes-sufficient baseline in the conjugate Gaussian model: for $\phi_{\text{suff}}(\Pi_k) = (X, m_k, \Sigma_k)$, the representation deficiency vanishes exactly, $A_k(\phi_{\text{suff}}) = 0$, so the full excess risk coincides with the readout-mismatch term $C_k(q; \phi_{\text{suff}})$.

three exact quantities. Panel (a) plots G_k against ΔBG_k . Panel (b) shows how the first crossing point moves under a controlled level of out-of-distribution length degradation. Panel (c) switches to the conjugate Gaussian family and shows the Bayes-sufficient baseline ϕ_{suff} for which $A_k = 0$ exactly.

7.2 Conjugate hierarchical regression reference model

Let $w \sim \mathcal{N}(0, \tau^2 I_d)$, $X \sim \mathcal{N}(0, I_d)$, and $Y \mid X, w \sim \mathcal{N}(w^\top X, \sigma^2)$. Given $D_k = \{(X_i, Y_i)\}_{i=1}^k$, the posterior is $w \mid D_k \sim \mathcal{N}(m_k, \Sigma_k)$ with

$$\Sigma_k = \left(\tau^{-2} I_d + \sigma^{-2} \sum_{i=1}^k X_i X_i^\top \right)^{-1}, \quad m_k = \sigma^{-2} \Sigma_k \sum_{i=1}^k X_i Y_i,$$

and the Bayes predictive is

$$p^*(Y \mid X, D_k) = \mathcal{N}(m_k^\top X, \sigma^2 + X^\top \Sigma_k X).$$

Hence

$$H_k = \frac{1}{2} \mathbb{E} \left[\log(2\pi e(\sigma^2 + X^\top \Sigma_k X)) \right], \quad G_k = \frac{1}{2} \mathbb{E} \left[\log \frac{\sigma^2 + X^\top \Sigma_k X}{\sigma^2 + X^\top \Sigma_{k+1} X} \right].$$

If a learned predictor outputs $q_\theta(Y \mid \Pi_k) = \mathcal{N}(\hat{\mu}_k, \hat{v}_k)$, then

$$\text{BG}_k(q_\theta) = \mathbb{E} \left[\text{KL} \left(\mathcal{N}(\mu_k^*, v_k^*) \parallel \mathcal{N}(\hat{\mu}_k, \hat{v}_k) \right) \right],$$

with $(\mu_k^*, v_k^*) = (m_k^\top X, \sigma^2 + X^\top \Sigma_k X)$, so the exact onset criterion can again be evaluated without ambiguity.

This family also makes the representation split concrete. For the posterior-summary representation

$$\phi_{\text{suff}}(\Pi_k) := (X, m_k, \Sigma_k),$$

the predictive law depends on the full prompt only through (X, m_k, Σ_k) , so $A_k(\phi_{\text{suff}}) = 0$ exactly and the residual excess risk is pure readout mismatch:

$$R_k(q_\theta) - H(Y \mid S_k) = C_k(q_\theta; \phi_{\text{suff}}).$$

This Bayes-sufficient baseline is a simple reference case for the representation-split theorem. Geometry-changing prompt manipulations can discard Bayes-relevant information, amplify readout mismatch, or do both; the tractable reference model separates those effects analytically.

The reference models fix the interpretation of the empirical audit. In these models the turning point never comes from the Bayes term itself; it appears only when diminishing Bayes gains are overtaken by model-side instability. The representation split then separates representation loss from readout mismatch, which is the distinction later tested with held-out readouts on frozen LLM states.

8 Experimental evaluation

The experiments are organized as a reliability audit rather than as a benchmark sweep. The accounting law says what must explain a harmful log-loss step; the experiments ask which observable mechanisms expose and repair that failure mode in deployed LLMs. Two layers are tested. The first is prompt shift: amplifying prompt geometry while keeping the demonstration path fixed should expose harmful late- k regimes when geometry is a stressor, and averaging over invariant orderings should reduce order-induced mismatch. The second is repairability: readout recovery and held-out calibration splits should indicate whether late-shot log-loss harm can be corrected without changing the underlying representation or model. ARC-Challenge candidate scoring is the primary setting because the prediction target is fixed and NLL, entropy, and perturbation disagreement come from the same cached distributions. The main evidence consists of the ARC audit case, order ensembling at $k=16$, held-out readout recovery, and the HellaSwag calibration-policy comparison against Fixed@16. Additional held-out and official-split checks are reported as appendix stress tests for the operating range of these conclusions.

Box 1. Offline diagnostic procedure for shot-budget selection. Given a model q_θ , task family, prompt template, candidate k -grid, and Bayes-invariant perturbation family, an audit proceeds as follows:

1. evaluate candidate distributions over the logged k -grid;
2. compute NLL where labels are available and predictive entropy as a gain-side proxy;
3. compute perturbation disagreement as a mismatch-side proxy;
4. flag regimes where entropy drops flatten while disagreement rises;
5. validate a conservative fixed k , entropy rule, or entropy-plus-disagreement rule on held-out logged curves.

Steps 1–3 require scoring every grid point and m perturbations per query, for $O(|\text{grid}| \cdot (1 + m))$ forward passes per logged query. The output is a range of potentially reliable shot budgets, diagnostic curves, and a candidate value of k for further validation. The protocol is designed for offline or periodic analysis; online deployment would require additional sequential-search accounting.

8.1 Cross-task accuracy checks

Table 1 Cross-task accuracy summary. ARC-Easy, ARC-Challenge, and BBH-Salient rows use the corresponding ManyICLBench prompt variants (Zou et al. 2025); 40/rep denotes 40 evaluation queries per replicate. The $\pm$ term is the standard deviation across the available replicates (ManyICLBench `seed0/seed1/seed2` splits or HellaSwag seeds). Each row reports peak mean accuracy, the corresponding k , and the worst post-peak drop. Dagger marks rows whose worst drop occurs at the archived $k = 32$ endpoint; parentheses give the corresponding drop on the controlled $k \leq 16$ grid. These rows are breadth checks; the log-loss mechanism analysis remains centered on ARC-Challenge candidate scoring.

Model	Setting	Peak mean acc.	Peak k	Worst drop (pp)
Qwen2.5-3B	HellaSwag (Order; 40/rep)	65.8 ± 11.8	4	16.7
Qwen2.5-3B	ARC-Easy (Order; 2k)	92.5 ± 2.5	8	$13.2^\dagger$ (1.7)
Qwen2.5-3B	ARC-Challenge (Order; 8k)	80.7 ± 2.4	0	$8.1^\dagger$ (2.5)
Qwen2.5-3B	BBH-Salient (Order)	56.7 ± 2.9	4	$11.7^\dagger$ (9.2)
OLMo-3-7B	HellaSwag (Order; 40/rep)	54.2 ± 6.3	4	30.8
OLMo-3-7B	ARC-Easy (Order; 2k)	91.7 ± 2.9	4	$58.3^\dagger$ (10.9)
OLMo-3-7B	ARC-Challenge (Order; 8k)	72.5 ± 1.3	2	$48.0^\dagger$ (12.3)
OLMo-3-7B	BBH-Salient (Order)	50.0 ± 6.6	1	$40.0^\dagger$ (37.5)
Mistral-7B	HellaSwag (Order; 40/rep)	54.8 ± 1.0	8	6.0
Mistral-7B	ARC-Easy (Order; 2k)	78.3 ± 5.8	2, 4	$7.3^\dagger$ (3.3)
Mistral-7B	ARC-Challenge (Order; 8k)	67.2 ± 2.1	1	$6.4^\dagger$ (1.4)
Mistral-7B	BBH-Salient (Order)	52.5 ± 0.0	0	5.0

Table 1 gives breadth checks around the ARC mechanism study. It condenses smaller-scale accuracy protocols, including 40-query-per-replicate HellaSwag, ARC-Easy, and BBH-Salient rows, together with the full 200-query ARC-Challenge split. The table shows that shot-budget sensitivity appears outside the primary ARC setting, but also that the size and direction of the effect depend on model, task, prompt family, and endpoint. Parenthesized $k \leq 16$ values are the ones comparable to the mechanism analyses; the OLMo $k = 32$ ARC-Challenge cells are degenerate because their mean accuracy falls below the four-way uniform-guessing rate (Supplementary

Appendix F.3). These rows motivate the log-loss audit without replacing it. The candidate-scoring analysis remains centered on ARC-Challenge, while Section 8.4 uses HellaSwag to test held-out policy selection on disjoint examples.

Supplementary Appendix F.2 reruns fixed 200-query ARC subsets on ManyICLBench `seed0/seed1/seed2`. Under perturbation-averaged Order prompts, Qwen has mean worst drops of 4.1 ± 1.4 pp (1k) and 3.3 ± 1.9 pp (8k), while Mistral has 2.5 ± 2.8 pp (1k) and 3.3 ± 1.1 pp (8k). Template-family reruns are less stable, especially for Qwen at 8k. We therefore treat Order and Pad perturbations as the controlled prompt families for the main mechanism analysis, and use broader template rewrites as stress checks.

The supplementary cross-task checks on HellaSwag, ARC-Easy, and BBH-Salient (Supplementary Appendix F.12) reinforce the same design lesson: shot budgets must be audited on the deployed prompt path. The mixed pattern across tasks and models is the reason the paper reports a diagnostic framework rather than a global many-shot rule.

8.2 ARC-Challenge log-loss analysis

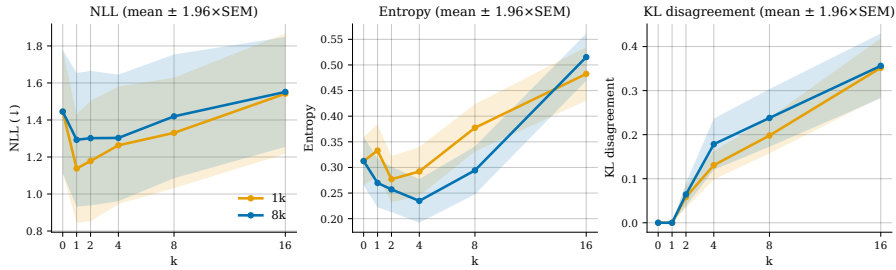


Fig. 3 ARC-Challenge (ManyICLBench `seed0`; fixed 200-query evaluation subset; **Qwen2.5-3B-Instruct**) baseline curves comparing 1k vs. 8k prompt variants. **Left:** NLL on the baseline nested prompt. **Middle:** predictive entropy on the same prompt. **Right:** KL-based disagreement score under the **Order** family. The descriptive signature is an interior NLL minimum followed by increased disagreement at larger k .

We evaluate **Qwen2.5-3B-Instruct**, **OLMo-3-7B-Instruct**, and **Mistral-7B-Instruct-v0.3** (shortened to Qwen2.5-3B, OLMo-3-7B, and Mistral-7B in tables) on ARC-Challenge using candidate scoring over $\{A, B, C, D\}$ along a nested grid $k \in \{0, 1, 2, 4, 8, 16\}$, evaluating a fixed 200-query subset per setting. The labels 1k, 2k, and 8k are ManyICLBench prompt variants: each variant provides a prompt prefix containing a pool of demonstrations, and we form nested prompts by taking the first k demonstrations from that pool; the larger variants support deeper k grids. NLL is the negative log-probability of the gold letter under the model distribution renormalized over the four candidates; hence $H_k \leq \log 4$, which makes the observable floor in Remark 3 valid. Qwen and OLMo are the two primary families for the ARC analysis (Figs. 3 and 4); Mistral is reported as a companion family in Table 3 and Supplementary Appendix F.13. Unless stated otherwise, the main curves use ManyICLBench

seed0. The primary diagnostics are NLL on the baseline prompt, predictive entropy, and Bayes-invariant disagreement under strong perturbation families, mainly **Order** and **Pad**. ARC is the primary diagnostic setting because candidate scoring keeps the target quantity fixed while all three observables are measured on the same cached runs.

Fig. 3 shows the ARC audit case: NLL reaches an interior minimum and then rises while disagreement increases. Accuracy summaries are secondary readouts of the same deterioration. Fig. 4 shows adjacent-grid traces for the two primary families. Harmful coarse steps in ΔNLL tend to occur where entropy drops weaken and disagreement growth rises; Theorem 2 and Corollary 4 make this relation exact for the latent terms $(G_k, \Delta\text{BG}_k)$, while Supplementary Appendix I measures how much of that structure appears in the observable proxies. The strict proxy audit gives an important design lesson: disagreement is valuable as a mismatch cross-check and intervention target, but entropy and confidence-drop baselines remain strong warning features for step-level prediction.

The candidate-scoring floor gives one direct certificate. Whenever the NLL curve exceeds $\log 4 \approx 1.386$ nats, Remark 3 certifies a nonzero Bayes gap with no auxiliary assumptions. On the **seed0** subset, OLMo-3-7B-Instruct’s $k=16$ NLL of 1.552 under the 8k Order setting (Supplementary Appendix F) gives $\text{BG}_{16}(q_\theta) \geq 0.166$ nats. For Qwen and Mistral the floor is inactive, so the audit relies on perturbations, readouts, and held-out calibration.

Stress-tested scope. The ARC case above is used as a controlled audit case. Supplementary Appendix F.4 and Supplementary Appendix F.6 report additional split and protocol stress tests, which document where the same late-step shape transfers and where it does not. These checks are useful because they keep the empirical story tied to a concrete prompt path. The main positive signals in the audit are the label-space Bayes-gap floor when it binds, order ensembling (Section 8.3, Table 3, and Table S9), and held-out readout recovery (Table 2).

Calibration and readout are the main repair channels. Where late-shot NLL harm appears, the strongest repairs are readout- and calibration-based. A held-out linear readout on frozen late- k representations recovers 0.96–1.58 nats of log-loss relative to the model’s own readout (Table 2B). Held-out ARC calibration removes most of Qwen’s small late-step rise (Supplementary Appendix F.4), and the HellaSwag audit below shows that log-loss-aware budget selection can improve over a fixed high-shot default. The common reading is that measured late-shot log-loss harm often lives in the readout/calibration layer of a representation that still contains usable task signal.

8.3 Mechanism checks via prompt interventions

The main mechanism analysis uses a matched 2×2 ARC design crossing prompt geometry with demonstration-path quality, paired with a held-out readout diagnostic on the same setting. Table 2 reports both analyses, with paired-bootstrap intervals for the factorial effects and for the held-out readout-recovery margins. The factorial results show a shared signal, but the dominant effect differs by model family: Qwen concentrates the effect in late- k $\Delta\text{NLL}(16-8)$, while OLMo concentrates it in disagreement growth under path degradation. The held-out readout results follow the same pattern. The probe protocol is summarized in Supplementary Appendix F.3; because

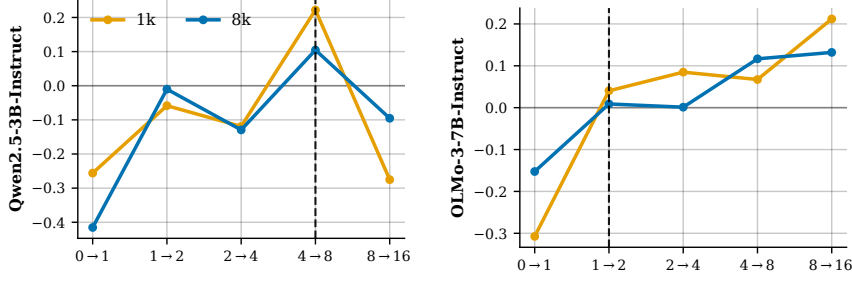


Fig. 4 ARC-Challenge (ManyICLBench seed0; fixed 200-query evaluation subset; Order family) adjacent-grid NLL changes across the two primary model families and 1k/8k prompt scales. ΔNLL is the exact risk-change quantity on the baseline prompt. Harmful coarse grid steps imply that a one-step gain-gap turning point occurred somewhere inside the interval. The entropy-drop and disagreement-growth panels from the same run are reported in Supplementary Appendix I, Fig. S1.

both readouts factor through the same frozen hidden states, Lemma 7 applies and the probe’s held-out advantage measures readout mismatch only. Both families show readout gains. Qwen has the sharper late- k NLL recovery margin, whereas OLMo shows recovery across more k values with a flatter 16–8 difference.

Table 2 Mechanism summaries on ARC-Challenge (ManyICLBench; 8k Order family). **Panel A** reports paired-bootstrap 95% CIs for the matched 2×2 geometry $\times$ path study on the late- k summaries $\Delta\text{NLL}(16-8)$ and $\Delta\hat{D}(16-8)$. **Panel B** reports the held-out readout analysis on seed2 examples, including paired-bootstrap 95% CIs for late and mean NLL recovery.

Panel A: Cross-family factorial summary

Model	Channel	Geom. effect	Path effect	Inter.
Qwen	ΔNLL	+0.141 [+0.028, +0.256]	-0.122 [-0.229, -0.027]	+0.183 [+0.017, +0.350]
	$\Delta\hat{D}$	-0.038 [-0.064, -0.012]	-0.007 [-0.031, +0.017]	+0.030 [-0.017, +0.080]
OLMo	ΔNLL	-0.087 [-0.197, +0.029]	-0.066 [-0.171, +0.032]	+0.020 [-0.132, +0.185]
	$\Delta\hat{D}$	+0.005 [-0.013, +0.024]	+0.038 [+0.021, +0.058]	-0.039 [-0.073, -0.008]

Panel B: Cross-family held-out readout summary

Model	Base ΔNLL	Probe ΔNLL	Late rec.	Mean NLL rec.
Qwen	+0.139	-0.042	+0.182 [+0.076, +0.292]	+0.958 [+0.690, +1.253]
OLMo	-0.038	+0.002	-0.040 [-0.132, +0.048]	+1.577 [+1.337, +1.830]

Reading. In Panel A, the larger effect for Qwen is associated with geometry-sensitive late- k ΔNLL , whereas the larger effect for OLMo is associated with path-sensitive KL-disagreement growth. Panel B shows recovery from a held-out readout in both families, with sharper late- k concentration for Qwen.

Under fixed hidden states Z_k , a held-out probe cannot recover Bayes-relevant information that is already absent from the representation: it cannot reduce the representation-deficiency term A_k . Any held-out log-loss improvement must come from reducing the readout-mismatch term C_k while leaving the representation unchanged. The held-out readout experiment suggests that part of the late- k deterioration lies

Table 3 Effect of order ensembling on ARC-Challenge (ManyICLBench `seed0`; fixed 200-query evaluation subset; 8k Order family; $n = 200$ queries; ensemble effect measured at $k=16$). The ensemble averages candidate distributions over the base prompt, one reverse-order prompt, and four shuffled-order prompts. Accuracy is reported as ensemble-minus-base percentage-point gain; NLL and disagreement are reported as base-minus-ensemble reductions. Brackets give query-paired bootstrap 95% intervals with $B = 10,000$ resamples. Full base and ensemble values, plus an independent held-out ARC replicate, are in Supplementary Appendix F.9.

Model	Acc. gain pp [95% CI]	NLL red. [95% CI]	Disagr. red. [95% CI]
Qwen2.5-3B	+2.0 [−0.5, +5.0]	0.031 [−0.023, 0.084]	0.090 [0.063, 0.120]
OLMo-3-7B	+3.0 [−1.0, +7.0]	0.212 [0.109, 0.325]	0.199 [0.145, 0.267]
Mistral-7B	+0.5 [+0.0, +1.5]	0.099 [0.037, 0.168]	0.018 [0.011, 0.027]

in the readout, not just in the representation. That is the distinction made by the representation-split theory.

Two prompt interventions are reported here. Position lock isolates query-start drift by padding prompts so that the query begins at a fixed absolute position while demonstrations and candidate scoring remain unchanged. On the archived 8k Pad run, its effect is family-dependent: Qwen’s late step reverses under the lock, whereas OLMo retains essentially all of its baseline late-step deterioration and has higher NLL at $k=16$ (Table S4). Query position is therefore one component of the prompt-shift mechanism, but controlling that component alone does not repair every family. Shift amplification tests the opposite direction by inserting delimiter padding before the query while keeping the same demonstrations and k -path. On the cached ManyICLBench `seed0` runs, this intervention moves the late- k regime in the predicted harmful direction for both primary families: the $k=16$ NLL effect is resolved for Qwen, and the two-family late-step average is positive, although the individual late-step intervals overlap zero (Fig. 5). Supplementary Appendix F.6 reports the official-split and padding-content stress tests, which show that delimiter padding is a protocol-dependent stressor rather than a universally reliable intervention. The more stable intervention is order ensembling, which reduces order-induced disagreement by averaging candidate distributions across invariant prompt orderings. Table 3 reports the compact cross-model effect, while Supplementary Appendix F.9 gives the full base and ensemble values and a rerun reconstruction from preserved per-example candidate distributions.

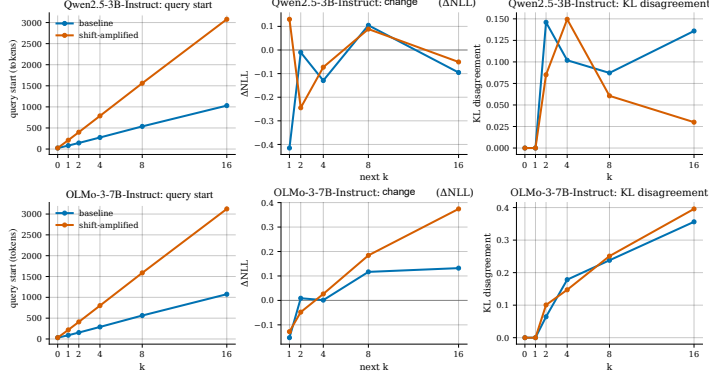


Fig. 5 ARC-Challenge shift-amplification diagnostic on the cached ManyICLBench *seed0* runs (8k Order family, $n = 200$). Delimiter padding is inserted before the query while demonstrations and the k -path are fixed. The intervention raises the two-family late-step $\Delta\text{NLL}(16-8)$ average by $+0.185$ $[0.035, 0.333]$; individual model intervals overlap zero, and Supplementary Appendix F.6 reports split and padding-content stress tests.

Statistical reporting. Intervals are query-paired bootstrap 95% intervals for the stated contrasts. The conclusions rely on the predeclared summaries in Table 2, Fig. 5, Table 3, and the held-out audit in Table 4; proxy traces are reported with their step-level warning value in Supplementary Appendix I. The replication archive includes the cached runs and scripts used for all bootstrap and calibration summaries.

8.4 Held-out policy audits from logged curves

After the mechanism checks, logged curves are used to ask whether calibration can choose a reliable shot policy before deployment. The audit is strictly offline: it reuses cached candidate distributions, uses labels only on a calibration split to choose among predeclared policies, and evaluates the chosen policy once on a disjoint test split. The result is a model-specific policy-selection check for the same log-loss/resource tradeoff studied above; online use would require a separate sequential decision rule.

The held-out HellaSwag audit uses demonstrations from the official training split and splits validation into 500 calibration examples and 1000 test examples with seed 20260609. Each example is scored as an explicit four-way multiple-choice prompt over the full endings. The calibration stage chooses among fixed budgets and locked uncertainty selectors on $k \in \{0, 1, 2, 4, 8, 16\}$ —per example, MaxEntropy picks the k with the largest predictive entropy and MinMargin the k with the smallest top-two probability margin—using NLL as the primary criterion, and evaluates the chosen policy once against Fixed@16. Supplementary Appendix F.10 gives selector definitions, accounting details, and the relation to earlier HellaSwag logged curves.

The selected policy is model-specific: Qwen chooses MinMargin, OLMo chooses Fixed@4, and Mistral chooses MaxEntropy. Qwen and OLMo improve held-out NLL over Fixed@16 with intervals excluding zero; Mistral’s NLL reduction is smaller and unresolved. Accuracy drops for Qwen and Mistral, while OLMo’s large accuracy gain mainly reflects avoiding a collapsed Fixed@16 regime near the guessing rate. This is

Table 4 Held-out HellaSwag calibration audit. A 500-example calibration split selects a model-specific policy from fixed budgets and locked uncertainty selectors; a disjoint 1000-example split evaluates the chosen policy against Fixed@16. Negative ΔNLL is better. Accuracy is secondary and is reported as a percentage-point change. Selector definitions, scoring protocol, and the locking procedure are given in Supplementary Appendix F.10; absolute accuracies should not be compared to the earlier ManyICLBench HellaSwag logged-curve tables.

Model	Selected policy	NLL	ΔNLL [95% CI]	$\Delta\text{acc.}$	Mean k	Tok. saved
Qwen2.5-3B	MinMargin	1.470	-0.312 [-0.452, -0.178]	-3.0 pp	10.69	30.7%
OLMo-3-7B	Fixed@4	1.344	-0.523 [-0.651, -0.391]	+30.2 pp	4.00	70.7%
Mistral-7B	MaxEntropy	0.854	-0.033 [-0.085, +0.020]	-2.4 pp	11.40	26.7%

the expected output of a log-loss audit: the selected rule is judged by the deployment score and by the system’s tolerance for an accuracy–NLL tradeoff. Supplementary Appendix F.10 adds a temperature-scaling reference, which confirms that calibration is a strong baseline and an important part of the repair channel. The main value of the audit is to expose that channel and to select a budgeted policy on a disjoint split, rather than to replace all post-hoc calibration methods.

9 Discussion

The paper establishes a local accounting rule for context growth under log-loss. Along a nested path, the Bayes predictor improves monotonically; a deployed predictor becomes worse only when Bayes-gap growth overtakes the remaining Bayes gain. This turns harmful shot scaling into an attribution problem rather than a generic long-context pathology. The representation split then gives the next question to ask: did the prompt representation lose task-relevant information, or did the readout become mismatched to information still present in the representation?

The empirical results support an audit discipline built around that question. The ARC case demonstrates the failure mode and gives a direct Bayes-gap floor at a high-NLL endpoint. Order ensembling reduces order-induced mismatch, held-out readouts recover substantial log-loss from frozen representations, and model-specific HellaSwag policies selected on a disjoint calibration split improve over a fixed high-shot default for two of three models. Supplementary stress tests document the operating range across splits and prompt protocols. The central deployment lesson remains: late-shot log-loss harm, when it appears, is often repairable at the readout or calibration layer.

The structural scope is fixed by Corollary 3: nested context expansion, fixed prediction target, and log-loss. Many-shot ICL and passive context-budget expansions such as retrieved evidence, memory records, and tool-trace summaries fit this object. Policies that replace earlier context, tools that change the environment, best-of- N output selection, and learned heuristic generation add decision rules beyond the one-step predictive-risk accounting. For systems that do follow a nested context path, the prescription is straightforward: treat the shot budget as a reliability variable and validate it under the deployment loss, rather than assuming that more context is automatically safer.

10 Conclusion

Nested-shot ICL exposes a basic deployment tradeoff: additional context can carry useful information while also moving the prompt distribution into a region where the model’s readout is worse matched to the target conditional law. The gain-gap identity makes that tradeoff exact under log-loss. The experiments then show how to audit it: measure adjacent NLL, inspect entropy and perturbation sensitivity, test invariant-order averaging, probe readout recoverability, and validate candidate budgets on held-out logged curves. This is the role of shot-budget reliability analysis in long-context LLM systems.

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Competing interests

The author declares no competing interests.

Ethics approval and consent to participate

Not applicable; this study does not report experiments involving human subjects.

Data availability

The experiments use publicly available benchmark datasets, including ARC, Hel-laSwag, and BBH via ManyICLBench, and open model checkpoints. Evaluation subsets, prompt paths, perturbation families, shot grids, and diagnostic quantities are documented in the manuscript and supplementary information.

Code availability

Online Resource 2 is the replication archive accompanying this submission. It contains manuscript sources, prompt configurations and table assets, derived figure files, ARC intervention caches, cached per-example candidate distributions for held-out audits, hidden-state readout-probe artifacts, and scripts for calibration and bootstrap analyses. The archive checksum is provided in a separate SHA-256 file.

Authors’ contributions

Junxin Fan was responsible for conceptualization, formal analysis, experimentation, software, visualization, writing, and revision of the manuscript.

Use of large language models

Large language models were used as writing and engineering assistants for copy editing, restructuring suggestions, LaTeX maintenance, literature-organization support, and code/log auditing. The author reviewed and approved the scientific claims, experiments, analysis, and final manuscript text.

Supplementary information

Online Resource 1 (Supplementary Information) contains limitations and scope details, technical proofs, toy constructions, the one-sided accuracy bridge, additional experimental protocols and tables, model-side surrogate audits, the theory-to-practice dictionary, and shot-selection heuristic details. The main text refers to its detailed sections as the Supplementary Appendix and to its extended tables as Supplementary Tables. Online Resource 2 is the replication archive and checksum described under Code availability.

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